

Article

Techno-Economic Optimization of a Grid-Tied PV/Battery System in Johannesburg's Subtropical Highland Climate

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Abstract

With rising energy costs and the need for sustainable power solutions in urban South African settings, grid-tied renewable energy systems have become viable alternatives for reducing dependence on traditional grid supply. This study investigates the techno-economic feasibility of a grid-connected hybrid photovoltaic (PV) and battery storage system designed for a commercial facility located in Johannesburg, South Africa—an area characterized by a subtropical highland climate. We conducted the analysis using the HOMER Grid software and evaluated the performance of the proposed PV/battery system against the baseline grid-only configuration. Simulation results indicate that the optimal systems, comprising 337 kW of flat-plate PV and 901 kWh of lithium-ion battery storage, offers a significant reduction in electricity expenditure, lowering the annual utility cost from \$39,229 to \$897. The system demonstrates a simple payback period of less than two years and achieves a net present value (NPV) of approximately \$449,491 over a 25-year project lifespan. In addition to delivering substantial cost savings, the proposed configuration also enhances energy resilience. Sensitivity analyses were conducted to assess the impact of variables such as inflation rate, discount rate, and load profile fluctuations on system performance and economic returns. The results affirm the suitability of hybrid grid-tied PV/battery systems for cost-effective, sustainable urban energy solutions in climates with high solar potential.



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Keywords: grid-tied PV system; battery energy storage; techno-economic analysis; HOMER Grid; subtropical highland climate

1. Introduction

South Africa's electricity sector has undergone a series of structural reforms and policy shifts in response to the country's persistent energy challenges, including supply shortages, aging infrastructure, and rising electricity tariffs. The unbundling of Eskom and the introduction of renewable energy procurement programs such as REIPPPP (Renewable Energy Independent Power Producer Procurement Programme) have opened opportunities for private sector participation and the integration of distributed energy resources [1]. However, despite these initiatives, urban commercial and residential consumers continue to face high electricity costs, periodic load shedding, and increasing uncertainty around energy reliability.

Urban facilities, particularly in cities like Johannesburg, are heavily dependent on grid electricity and are vulnerable to disruptions and escalating energy expenditures. In response, there is growing interest in adopting distributed renewable energy systems—especially grid-tied photovoltaic (PV) systems coupled with battery storage—as a cost-

effective and resilient alternative to conventional grid reliance [2]. Unlike rural electrification projects, which often focus on off-grid solutions to address access gaps, urban distributed energy systems prioritize economic optimization, peak shaving, and resilience to grid instability [3].

Given Johannesburg's favorable solar resource under its subtropical highland climate, hybrid PV and battery systems offer a promising pathway to reduce energy costs and mitigate load-shedding impacts. This challenge is further compounded by the escalating cost of electricity and the limited progress in improving grid reliability within urban centers. While large-scale grid expansion is less of a concern in urban areas, the volatility of energy pricing, aging infrastructure, and recurring load shedding have created an urgent need for distributed, consumer-level generation solutions. Grid-tied small-scale renewable energy systems, particularly those incorporating solar PV and battery storage, have emerged as viable strategies not only to reduce electricity costs but also to enhance resilience and energy independence. These systems serve as foundational elements in the broader transition toward decentralized, low-carbon energy networks. For commercial and residential consumers alike, such systems represent a strategic response to the financial and operational uncertainties of conventional grid dependence, particularly in cities like Johannesburg where energy reliability and affordability are critical to sustainable economic activity and urban development [4].

In South Africa, approximately 80% of electricity generation is derived from coal-fired thermal power plants, highlighting a strong dependency on fossil fuels for meeting national energy demands [5]. This reliance comes with high operational and maintenance costs, along with significant carbon dioxide (CO₂) and pollutant emissions, which contribute to climate change and pose health risks in densely populated urban areas. These challenges underscore the urgent need to diversify the energy mix by incorporating cleaner and more sustainable resources such as solar and wind energy. South Africa benefits from some of the world's highest solar irradiance levels, with annual solar energy availability far exceeding national consumption needs, making photovoltaic (PV) systems particularly well-suited for distributed generation across both rural and urban environments [6]. While utility-scale renewable deployment has gained traction through programs like REIPPPP, there remains substantial untapped potential for grid-tied distributed energy solutions at the facility level. As international agreements such as the Paris Climate Accord emphasize the reduction of greenhouse gas (GHG) emissions, South Africa has also committed to lowering its carbon intensity and transitioning toward a low-carbon economy [7,8]. Moreover, the rising cost of grid electricity, combined with the socioeconomic need for reliable power to sustain commercial activity and economic growth, supports a growing role for distributed renewable energy systems. These systems not only reduce emissions but also offer an economically attractive pathway to long-term energy resilience and affordability [9].

Furthermore, renewable energy (RE) technologies are becoming increasingly attractive for electricity generation in urban and peri-urban areas, not only due to their environmental benefits—such as reduced greenhouse gas (GHG) emissions and lower pollution—but also because they leverage abundant, naturally available energy sources like solar radiation. Numerous studies have been conducted globally to assess the feasibility and economic performance of hybrid systems combining renewable sources with conventional grid or diesel-based systems across different load profiles and settlement types. These investigations consistently demonstrate that such hybrid configurations enhance sustainability, reduce emissions, and offer long-term cost advantages despite higher initial capital costs [10–17]. Recent declines in the cost of renewable energy technologies (RETs), particularly in solar photovoltaics and battery storage, have made these systems increasingly cost-competitive, with a growing share of RET capacity now delivering electricity at a lower levelized cost

than conventional fossil-based generation options [18,19]. For instance, case studies in sub-Saharan Africa have shown significant savings when diesel-only systems are replaced with PV or PV/battery hybrids, with energy cost reductions of up to 30%, depending on local conditions [20]. Similarly, techno-economic analyses of decentralized PV systems have demonstrated that the viability of such systems is strongly site-dependent, influenced by solar irradiance, load profiles, and tariff structures. In one such study, an optimized PV/battery system delivered a net present value of \$449,491 and reduced the unit cost of energy from \$0.62/kWh (diesel) to \$0.04/kWh, confirming the economic advantage of distributed renewable systems in suitable urban settings [21].

1.1. Literature Review

In ref. [22], the viability of a rooftop solar PV system integrated with battery storage for supporting refrigeration and energy demands in an urban retail complex was assessed, demonstrating that distributed renewable systems can provide both functional and economic benefits within commercial precincts. The study found that such systems not only reduce energy costs but also enhance service continuity during grid outages. Similarly, the application of hybrid energy systems (HESs) in semi-urban settings was explored in [23], where various configurations combining solar, storage, and diesel backup were analyzed. The results confirmed that hybrid PV/battery systems provided superior cost-effectiveness and environmental performance compared to standalone grid or fossil-fuel-based alternatives. A study conducted in Riyadh, Saudi Arabia, also examined hybrid system configurations for commercial energy users, highlighting the region's strong solar potential. It found that, although wind contributed to total energy supply, its cost per unit (\$0.149/kWh) was significantly higher than that of solar energy (\$0.0637/kWh), reinforcing the economic superiority of PV systems in high-irradiance locations [24]. An integrated analysis of PV/battery/diesel configurations for an educational institution in Kaduna, Nigeria, presented in [25], concluded that the PV-dominated hybrid system offered substantial savings in electricity costs while minimizing greenhouse gas emissions. In a comparable urban study [12], the deployment of a rooftop PV/battery solution in a district of Dhaka demonstrated a 62% reduction in CO₂ emissions relative to the grid-only configuration, and even greater reductions when compared to diesel-dependent systems. In ref. [26], HOMER was used to optimize a PV/battery system for a small urban hospital in Giri town, where the system delivered a minimum net present cost of \$1.01 million and annual GHG emissions of just 2889.36 kg—demonstrating a strong balance between economic feasibility and environmental benefit. Lastly, in ref. [27], a techno-economic assessment of a hybrid PV/battery/diesel configuration for a peri-urban community in Sri Lanka revealed that the optimal system produced electricity at \$0.34/kWh—substantially lower than local grid tariffs—emphasizing the growing competitiveness of decentralized renewable energy technologies in diverse regional contexts.

In addition, studies such as Vargas-Salgado et al. [13] and Yadav et al. [14] have validated HOMER-based modeling approaches for techno-economic analysis, while Kuru-kuri et al. [20] emphasized policy-aligned optimization strategies for hybrid systems in urban contexts. These contributions support the reliability of simulation outputs and highlight the relevance of tax structures and financial instruments in evaluating the long-term cost-effectiveness of distributed renewable energy systems.

This research aims to support the advancement and deployment of distributed grid-connected renewable energy systems by providing a techno-economic feasibility and performance analysis of a grid-tied photovoltaic (PV) and battery energy storage system (BESS) for a commercial facility located in Johannesburg, South Africa, under subtropical highland climate conditions. The simulation is conducted using real solar resource data and

local energy load profiles, with battery storage included to mitigate the intermittent nature of solar power and ensure reliability during periods of low irradiance or grid instability. Although direct validation against a physical pilot installation was not performed, the model uses real hourly solar irradiance and measured facility load data to closely reflect operational conditions and support practical feasibility.

We simulated and compared multiple system configurations using the HOMER Grid software (2018 version), with a sensitivity analysis performed to evaluate both the economic and operational performance of the hybrid system under variations in key parameters such as the real discount rate, solar irradiance, load fluctuations, and battery minimum state of charge (SOC). The study further examines the local solar energy potential and its integration into commercial-scale energy infrastructure, providing actionable insights aligned with South Africa's broader energy policy objectives and supporting the implementation of national targets to increase renewable energy penetration in the electricity mix. The outcomes are intended to guide the future adoption of grid-tied PV and storage systems in urban settings as part of the country's commitment to sustainable and economically viable energy solutions.

1.2. Contribution

This study contributes to the growing body of research on distributed renewable energy systems by presenting a detailed techno-economic analysis of a grid-tied PV and battery energy storage configuration for an urban commercial facility in Johannesburg, South Africa. The contributions are threefold:

1. It evaluates the cost-saving potential of hybrid PV/BESS systems in an urban, grid-connected context with real electricity tariffs and load profiles.
2. It applies HOMER Grid to simulate multiple configurations, offering a rigorous performance comparison across key economic and operational metrics such as payback period, net present value (NPV), and annual utility cost savings.
3. It provides a structured sensitivity analysis of key system parameters—including solar irradiance, inflation rate, and load demand fluctuations—giving insight into system robustness and investment risk under varying economic and environmental conditions.

1.3. Study Novelty

Unlike prior studies that focus predominantly on rural, off-grid electrification using combinations of solar, wind, hydro, and diesel, this study is novel in its focus on urban, grid-connected PV and battery systems under subtropical highland climate conditions, a context that is underrepresented in the current literature. Key elements of novelty include the following:

1. The use of real Johannesburg energy demand data and local solar resource conditions to optimize the system design.
2. A focus on urban commercial applications, where cost reduction, reliability, and peak shaving are critical, rather than access provision as in rural scenarios.
3. A demonstration of how distributed PV/battery systems can provide energy cost savings of over 97%, with a simple payback of less than one year—underscoring their economic competitiveness even without feed-in tariffs or large-scale subsidies.
4. An integrated analysis that aligns with South Africa's renewable energy policy objectives, filling a research gap at the intersection of urban infrastructure, solar potential, and distributed generation economics.

2. Materials and Methods

Selecting appropriate evaluation criteria is essential for accurately assessing the operational and economic performance of hybrid renewable energy systems, particularly in urban, grid-tied configurations. This study follows a structured assessment approach, which includes the following:

1. Geographic and climatic specifications of the selected urban location;
2. Characteristics of the proposed hybrid PV/battery system and facility load profile;
3. Design and component specifications;
4. Cost inputs and financial assumptions;
5. Simulation framework and sensitivity analysis carried out using the HOMER Grid software.

2.1. Potential Renewable Energy Resources in Johannesburg, South Africa

South Africa possesses a wealth of renewable energy resources, particularly solar, that are highly viable for integration into distributed generation systems. While hydropower has traditionally played a role in national-scale generation, it is solar energy that offers the most practical and scalable option for urban areas like Johannesburg, given the region's altitude, climate, and consistent irradiance levels. Located at an altitude of approximately 1762 m, with coordinates at latitude -26.20227° and longitude 28.04363° , Johannesburg benefits from its subtropical highland climate (Köppen: Cwb), characterized by clear skies during winter and moderate summer cloud cover. These climatic conditions allow for high solar availability throughout most of the year. According to hourly solar radiation forecasts (Figure 1a), Johannesburg can experience a total daily solar radiation value of approximately 3969 Wh/m^2 on a typical clear day, with peak irradiation values exceeding 600 W/m^2 between 11:00 and 14:00. This stable and predictable irradiance profile makes the city particularly suitable for PV generation. In contrast to rural regions where hydropower might be viable through small hydro schemes, Johannesburg's renewable energy potential is firmly anchored in solar resources. Although not typically associated with river-based small hydropower development due to limited surface water flow and topographic constraints, the city's renewable energy transition has embraced grid-tied solar PV systems, increasingly augmented by battery energy storage to manage intermittency and load balancing. As illustrated in the astronomical observation of the local sky (Figure 1b), clear sky conditions and optimal solar angles are prevalent, enabling effective PV capture. Additionally, the city's infrastructure and energy policy framework are progressively accommodating decentralized energy solutions to relieve pressure on the national grid and mitigate the impact of load shedding.

With a mean daily solar radiation range of 4.5 to $6.5 \text{ kWh/m}^2/\text{day}$, Johannesburg sits well within the global optimum range for photovoltaic applications [28]. This makes PV/battery hybrid systems not only technically feasible but also economically attractive, especially for urban commercial and industrial users seeking resilience and cost reduction. From the observed data and modeled conditions, it is evident that Johannesburg's solar profile supports the deployment of high-efficiency distributed PV systems, and, when coupled with battery storage, these systems can dramatically reduce grid dependency. This potential reinforces the strategic relevance of solar-focused hybrid systems in meeting both short- and long-term energy needs in the Gauteng region.

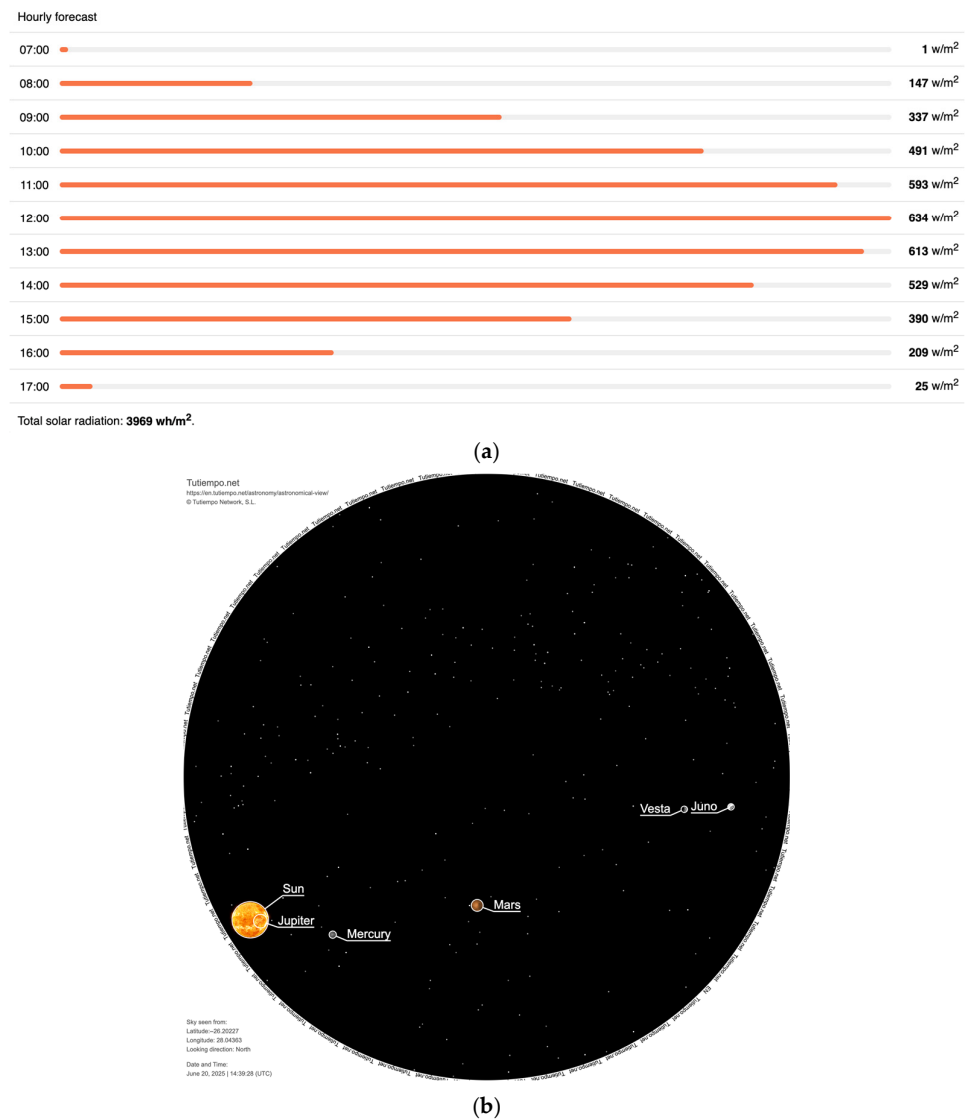


Figure 1. (a) Hourly solar radiation forecast for Johannesburg on a clear day. (b) Astronomical sky map from Johannesburg [28].

2.2. Specifications of the Selected Location

The selected site for this study is a commercial facility located at Melville, Johannesburg, within the Gauteng province of South Africa. The site lies approximately 5 km west of central Johannesburg and is situated at a latitude of -26.20227° , longitude 28.04363° , and an altitude of approximately 1762 m above sea level. The area experiences a sub-tropical highland climate (Cwb), which is favorable for solar energy harvesting due to consistent clear skies during winter and moderate summer cloudiness. Electricity at the site is supplied by the national grid under the RSA grid tariff structure, which is subject to price volatility and frequent disruptions due to scheduled load shedding. According to recent utility billing data, the facility's baseline energy consumption is approximately 839 kWh/day, with a peak demand of 178 kW, resulting in an annual electricity expenditure of approximately \$39,229. The reliability of the grid is a growing concern for businesses in this region, prompting the adoption of distributed, grid-tied PV systems coupled with battery energy storage to reduce costs and enhance resilience.

Unlike remote or rural areas with potential access to small hydropower resources, Johannesburg's energy strategy emphasizes urban solar PV deployment, supported by strong solar irradiation levels and favorable policy direction. The location receives a total

daily solar radiation of approximately 3969 Wh/m^2 , with peak values reaching more than 600 W/m^2 during midday hours. These conditions are optimal for photovoltaic generation and justify the selection of this site for PV/BESS system implementation. The PV system is assumed to be rooftop-mounted on the commercial facility; hence, no additional land-use costs were included in the economic assessment. This study does not include diesel backup generation, as the focus is on grid-connected solar and storage configurations that align with South Africa's renewable energy transition strategy and urban sustainability objectives. While diesel generators are often used to enhance reliability in cases of prolonged grid failure, they were excluded here to preserve a zero-emission system architecture. However, we acknowledge their value as a fallback in extreme outage scenarios and recognize that real-world implementations may consider diesel backup as a supplementary layer of energy security, depending on risk tolerance and regulatory frameworks.

The selection of this site reflects a broader interest in scalable, replicable solutions for commercial buildings seeking to reduce dependence on an aging and intermittently reliable electricity grid.

2.3. Proposed Hybrid Generation System and Load Profile Analysis

The proposed hybrid energy system designed for this study comprises three primary components: a grid-connected PV system, a lithium-ion battery storage system (LI ASM), and a bidirectional power converter to facilitate energy exchange between the AC and DC buses. The grid acts as both a backup and supplementary source to ensure supply reliability during periods of insufficient solar generation or high load demand. The configuration of this hybrid system is illustrated in Figure 2, with system control and optimization conducted using HOMER Grid software. The PV component is modeled as a flat-plate system with a nominal capacity of 337 kW, while the battery storage unit has an energy capacity of 901 kWh. The converter supports real-time energy management between sources, enabling the system to effectively dispatch energy depending on generation, load demand, and grid status. The facility under study is in Melville, Johannesburg, and experiences a typical daily electrical demand of 840.05 kWh/day with a peak demand of 178.43 kW. The battery storage system is installed indoors with adequate ventilation and is managed by a battery management system (BMS) to ensure thermal stability and fire safety under Johannesburg's moderate climatic conditions.

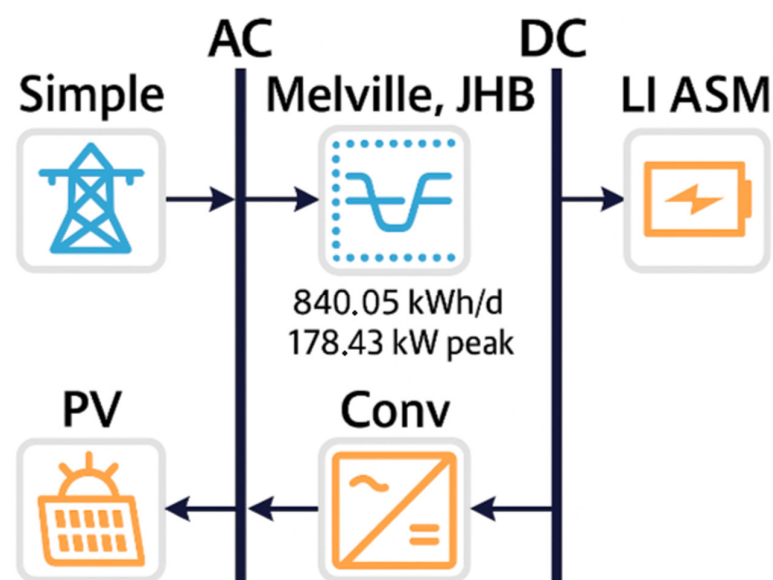


Figure 2. HOMER Grid schematic of the proposed hybrid system.

We modeled the energy consumption data as an hourly load profile over a full year to accurately capture diurnal and seasonal variations. Figure 3a presents the yearly heat map of demand, indicating elevated usage in the evening hours throughout the year, especially around 18:00 to 21:00. Figure 3b shows the average daily load profile, with noticeable peaks in the late afternoon and early evening corresponding to operational and lighting loads. Figure 3c illustrates the monthly/seasonal distribution, where energy consumption patterns remain relatively consistent across the year, with slightly higher usage observed during the winter months due to increased lighting and heating needs.

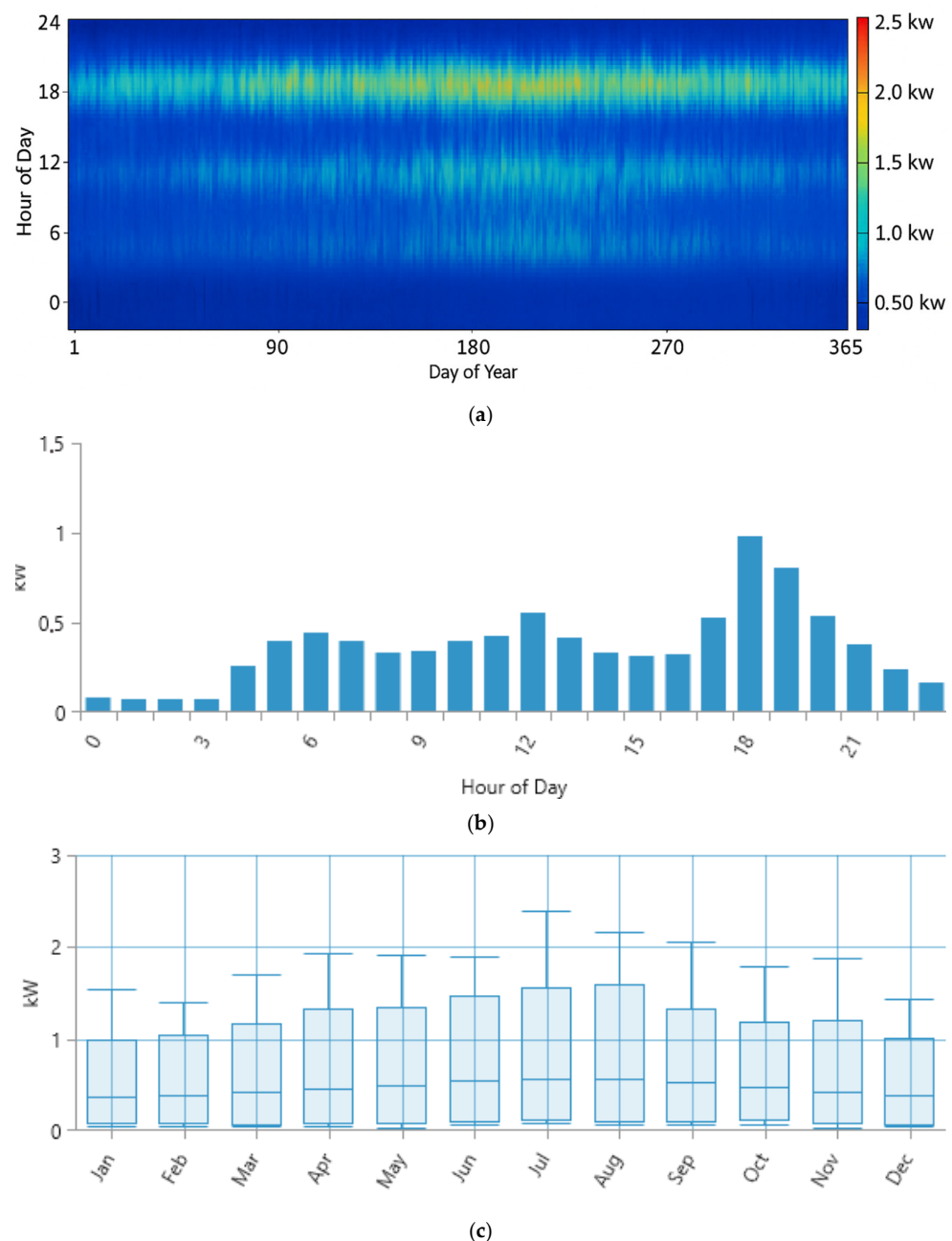


Figure 3. (a) Yearly load profile heatmap. (b) Average daily load. (c) Seasonal load variation by month.

The demand profile includes typical urban commercial loads such as lighting, cooling, refrigeration, and electronics. Most load activity occurs between 06:00 and 21:00, with the highest consumption observed during the early evening due to the concurrent operation of multiple systems. To simulate system flexibility and real-world uncertainty, HOMER Grid incorporated a 10% time-step and daily variability to reflect stochastic behavior in load and

solar resource inputs. The selected hybrid configuration was compared against alternative scenarios, including grid-only, PV-only, and PV plus battery without grid feedback, to assess optimality in terms of cost, energy balance, and reliability. Additionally, we performed sensitivity analyses by varying economic and technical parameters such as discount rate, battery depth of discharge, solar irradiance, and tariff escalation rate, enabling a robust understanding of system performance across different planning conditions.

To simulate realistic operating conditions, HOMER Grid applies a stochastic modeling approach by introducing a 10% time-step variability (random hourly deviations) and 10% daily variability (random day-to-day fluctuations), which reflect natural variations in load and solar inputs based on a Gaussian distribution framework. This enhances the robustness of the simulation by accounting for unpredictable user behavior and weather conditions.

2.4. Hybrid System Components and Costs

This study incorporates three primary components in the hybrid system: photovoltaic (PV) generation, lithium-ion battery storage, and a DC–AC converter. Component specifications and economic inputs are presented in Table 1. HOMER Grid was used to perform optimization simulations across a range of component sizes to determine the configuration that provides the best techno-economic performance for the site. Energy management was modeled using a load-following dispatch strategy, where the PV system supplies the load first, supported by the battery system, and the grid serves as a backup when renewable sources are insufficient. The project lifetime was assumed to be 25 years, with a real discount rate of 8% and 2% inflation rate. Reliability constraints were set by allowing 3% maximum annual capacity shortage, 10% operating reserve of load, and 25% solar output, aligning with industry-accepted reliability standards [29].

Table 1. System components and economic specifications.

Component	Parameter	Value
PV (Flat-Plate, Generic)	Rated Capacity	337 kW
	Annual Production	597,327 kWh
	Specific Yield	1774 kWh/kW
	Capital Cost	\$40,221
	Operations and Maintenance (O&M) Cost	\$34.26/year
	LCOE	\$0.0053/kWh
Battery (Li-Ion, Generic 1 kWh ASM)	PV Penetration	195%
	Rated Capacity	901 kWh
	Annual Throughput	167,117 kWh/year
	Capital Cost	\$21,711
	Maintenance Cost	\$399/year
	Expected Life	15 years
	Storage Losses	14,007 kWh/year
Converter (Generic)	Autonomy	18.0 h
	Rated Capacity	178 kW
	Energy In/Out	160,293 kWh/152,278 kWh
	Capital Cost	\$0.00 (included)
	Losses	8015 kWh/year
	Efficiency	~95%
Grid	Hours of Operation	5128 h/year
	Annual Energy Supplied (base case)	306,618 kWh
	Annual Energy Supplied (with PV/BESS)	7011 kWh
	Annual Utility Cost (base case)	\$39,229
	Annual Utility Cost (with PV/BESS)	\$897

Note: The PV O&M cost is based on HOMER Grid's default assumptions.

The grid electricity purchase rate was set at \$0.13/kWh, based on the official 2025 municipal tariff structure for Johannesburg. This flat tariff was selected to simplify the analysis; however, in practice, Johannesburg's commercial tariffs often include time-of-use pricing, fixed monthly charges, and demand-based components. These were not modeled due to limited resolution in publicly available billing data but are acknowledged as factors that may influence actual savings margins. Figure 4a–c illustrates the corresponding (a) PV: Generic flat-plate PV, (b) Storage: Generic 1 kWh Li-Ion [ASM], and (c) Converter: Generic large, free converter.

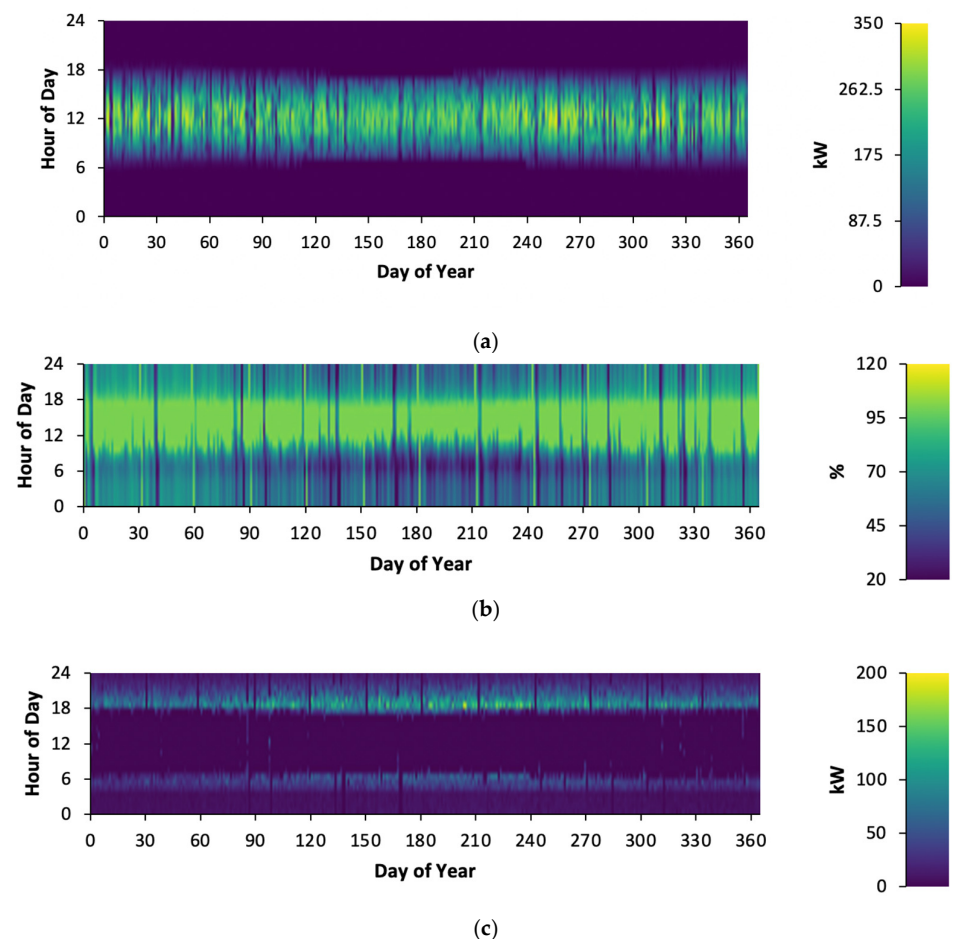


Figure 4. (a) PV: Generic flat-plate PV. (b) Storage: Generic 1 kWh Li-Ion [ASM]. (c) Converter: Generic large, free converter.

The financial viability was assessed under generic utility tariff structures applicable in Johannesburg. While the model includes generic tax benefits such as depreciation and investment credits, these are illustrative and not based on specific South African tax codes. A detailed country-specific financial policy assessment lies beyond the current study's scope and is recommended for future research to refine cost accuracy. Diesel-based options were excluded from the final model due to high emissions and operational costs, favoring a clean-energy urban application with zero fossil fuel dependency.

2.5. Mathematical Model

2.5.1. Modeling of a Grid-Tied PV and Battery Energy System

In this study, the modeling of a grid-connected hybrid PV/battery system was carried out using the HOMER Grid software, which performs techno-economic simulations by solving power balance equations under load-following dispatch strategies. The goal is to minimize total net present cost while ensuring reliability through energy storage and grid support. The

energy output from the PV system is modeled as a function of the solar irradiance incident on the panel surface, panel efficiency, and area, as expressed in Equation (1) [30,31].

$$P_{PV} = G_t \cdot A_{PV} \cdot \eta_{PV} \quad (1)$$

where

P_{PV} —instantaneous PV power output (kW);

G_t —global solar radiation at time t (kW/m²);

A_{PV} —area of PV array (m²);

η_{PV} —conversion efficiency of the PV modules.

The battery storage system is modeled using a state-of-charge (SOC) equation that updates dynamically based on charge/discharge cycles, efficiency, and power flow, as expressed in Equation (2) [30,31].

$$SOC_t = SOC_{t-1} + \frac{\eta_c \cdot P_{ch} \cdot \Delta t}{E_{max}} - \frac{P_{dis} \cdot \Delta t}{\eta_d \cdot E_{max}} \quad (2)$$

where

SOC_t —state of charge at time t ;

P_{ch}, P_{dis} —battery charge/discharge power (kW);

η_c, η_d —charge/discharge efficiency;

E_{max} —nominal capacity of the battery (kWh);

Δt —time interval (h).

The battery model also enforces constraints on minimum and maximum SOC levels to preserve battery life and ensure system resilience. In HOMER Grid, the power balance is calculated for each time step based on the following expression expressed in Equation (3) [30,31].

$$P_{load} = P_{PV} + P_{battery} + P_{grid} \quad (3)$$

where

P_{load} —facility electrical demand;

P_{PV} —solar PV contribution;

$P_{battery}$ —battery discharge (or negative during charging);

P_{grid} —grid import (positive) or export (negative).

The software solves these equations for each hour of the simulation year using actual solar radiation, load data, and component parameters to determine system behavior under different economic and technical conditions.

2.5.2. Modeling of a PV System and Temperature Effects

PV modules convert solar energy into electrical energy and are widely used for distributed generation applications, including residential, commercial, and industrial facilities. The output performance of a PV system is primarily influenced by solar irradiance and module temperature, both of which are dynamic in Johannesburg's subtropical highland climate. These effects are incorporated into HOMER Grid's simulation engine for accurate performance modeling. The output power of the PV array at any time step is estimated using the irradiance and temperature-adjusted model, as expressed in Equation (4) [30,31].

$$P_{PV} = Y_{PV} f_{PV} \left(\frac{G_T}{G_{T,STC}} \right) [1 + \alpha_p (T_C - T_{C,STC})] \quad (4)$$

where

P_{PV} —instantaneous output of the PV array (kW);
 Y_{PV} —PV array rated capacity under Standard Test Conditions (STC);
 f_{PV} —PV derating factor (%);
 G_T —global solar irradiance at the current time (kW/m²);
 $G_{T,STC}$ —irradiance at STC (1 kW/m²);
 α_p —temperature coefficient of power (%/°C);
 T_C —PV cell temperature (°C);
 $T_{C,STC}$ —cell temperature at STC (25 °C).

Since PV modules tend to lose efficiency at higher temperatures, a derating function is used to account for thermal impacts. The derating factor as a function of temperature is expressed in Equation (5) [30,31].

$$f_{temp} = \frac{1 + \alpha_p \left(\frac{T_{C,NOTC} - T_{a,NOTC}}{I_{T,NOTC}} \right) - T_{C,STC}}{1 + \alpha_p I_T \left(\frac{T_{C,NOTC} - T_{a,NOTC}}{I_{T,NOTC}} \right) \left(\frac{\eta_{mp,STC}}{0.9} \right)} \quad (5)$$

where

$T_{C,NOTC}$ —nominal operating cell temperature (°C);
 $T_{a,NOTC}$ —ambient temperature at NOCT (20 °C);
 $I_{T,NOTC}$ —incident irradiance at operating and NOCT levels (kW/m²);
 $\eta_{mp,STC}$ —PV module efficiency at STC;
 α_p —temperature coefficient (%/°C).

This formulation is particularly relevant for Johannesburg's climate, which exhibits moderate ambient temperatures but significant daily solar variability. Capturing the real-time irradiance and thermal response of the PV modules, the simulation ensures accurate prediction of energy yields and cost-performance of the hybrid system across all seasons.

2.5.3. Modeling of Economic Parameters

The economic viability of the proposed hybrid PV/battery energy system is assessed using two key financial indicators: the net present cost (NPC) and the Levelized Cost of Energy (LCOE). These metrics reflect the long-term cost-effectiveness of the system configuration over the project lifetime. The net present cost represents the present value of all costs incurred throughout the system's lifetime, including capital investment, replacement costs, operational and maintenance (O&M) costs, and any residual value at the end of the project. It is calculated as expressed in Equation (6) [30,31].

$$C_{NPC} = \frac{TAC}{CRF(i, n)} \quad (6)$$

where

C_{NPC} —net present cost in USD;
 TAC —total annualized cost (USD/year);
 $CRF(i, n)$ —capital recovery factor;
 i —real discount rate (fraction);
 n —project lifetime (years).

The capital recovery factor (CRF) is used to convert an annualized cost into a present value and is defined by Equation (7) [30,31]. The tax incentives applied in this study, including the Modified Accelerated Cost Recovery System (MACRS) and the Investment Tax Credit (ITC), are based on standard U.S. federal renewable energy financial frameworks, as supported in HOMER Grid's economic modeling defaults. While South Africa's incentive

structure differs, the use of these inputs reflects a best-case scenario for investment return modeling and aligns with practices in international feasibility studies. Future localized cost modeling could incorporate South African-specific depreciation schedules or subsidy schemes such as those proposed under Renewable Energy Independent Power Producer Procurement Programme (REIPPPP).

$$CRF(i, n) = \frac{i(1+i)^n}{(1+i)^n - 1} \quad (7)$$

This allows the economic comparison of systems with different operational lives and interest rate conditions. In this study, a 25-year lifespan and an 8% real discount rate were applied based on typical commercial investment assumptions for South Africa. The Levelized Cost of Energy (COE), expressed in USD per kilowatt-hour, represents the average cost of energy supplied over the system's operational period. It is calculated by dividing the total annualized cost by the total useful energy delivered, as expressed in Equation (8) [30,31].

$$COE = \frac{TAC}{E_{\text{served}}} \quad (8)$$

where

COE—cost of energy (\$/kWh);

E_{served} —total energy load served by the system annually (kWh).

These indicators enable a holistic understanding of the system's affordability and economic performance relative to grid-only supply or alternative configurations.

3. Results

In this analysis, the HOMER Grid simulation software was employed to identify the optimal grid-connected PV/battery system configuration for a commercial site in Johannesburg. The evaluation was based on a comprehensive techno-economic performance assessment using actual solar resource data and a real commercial load profile under subtropical highland climate conditions.

3.1. Optimization Results

This section presents the NPC and annualized financial breakdown for the proposed hybrid energy system composed of a 337 kW PV array, 901 kWh lithium-ion battery storage, and partial grid dependency, evaluated over a 25-year project horizon using HOMER Grid. The analysis incorporates capital costs, replacement, operational and maintenance (O&M), fuel, depreciation, tax incentives, and salvage values. Table 2 provides the total lifetime costs for the proposed system configuration.

Table 2. NPC project summary.

Component	Capital (\$)	Replacement (\$)	O&M (\$)	Salvage (\$)	Total (\$)
Bonus Depreciation	−99,384.07	0.00	0.00	0.00	−99,384.07
1 kWh Li-Ion Battery [ASM]	390,377.80	165,627.08	92,822.44	−31,172.69	617,654.63
Flat-Plate PV Array	723,169.21	0.00	7959.60	0.00	731,128.81
Investment Tax Credit	−334,064.10	0.00	0.00	0.00	−334,064.10
MACRS Depreciation	0.00	0.00	−84,989.47	0.00	−84,989.47
RSA Grid	0.00	0.00	208,473.66	0.00	208,473.66
System Total	680,098.84	165,627.08	224,266.23	−31,172.69	1,038,819.47

The lithium-ion battery bank accounts for a major share of lifetime cost at \$617,654.63, while the PV system contributes \$731,128.81. The grid dependency, although minimal, adds another \$208,473.66 to the overall expenditure. Incentives, particularly the Investment Tax Credit and MACRS depreciation, substantially reduce the net project cost. To evaluate economic feasibility on an annual basis, Table 3 tabulates the annualized cost breakdown. The total annualized cost is \$80,357.23, inclusive of all recurring financial obligations. The battery system incurs the highest recurring cost at \$47,778.29/year, while the PV contributes \$56,556.01/year. Tax incentives continue to offset costs significantly—with annual MACRS depreciation at −\$6574.31 and tax credits lowering annualized costs by −\$25,841.32.

Table 3. Annualized cost project summary.

Component	Capital (\$)	Replacement (\$)	O&M (\$)	Salvage (\$)	Total (\$)
Bonus Depreciation	−7687.79	0.00	0.00	0.00	−7687.79
1 kWh Li-Ion Battery [ASM]	30,197.43	12,811.98	7180.22	−2411.34	47,778.29
Flat-Plate PV Array	55,940.30	0.00	615.71	0.00	56,556.01
Investment Tax Credit	−25,841.32	0.00	0.00	0.00	−25,841.32
MACRS Depreciation	0.00	0.00	−6574.31	0.00	−6574.31
RSA Grid	0.00	0.00	16,126.35	0.00	16,126.35
System Total	52,608.62	12,811.98	17,347.97	−2411.34	80,357.23

The utility cost comparison in Figure 5 clearly illustrates the financial advantage of the proposed hybrid system over the base case (grid-only configuration). The base case system incurs utility bills ranging from approximately \$2500 to \$3900 per month, while the proposed system consistently maintains costs below \$300 per month—demonstrating a cost savings of more than 90% on average. This highlights the significant economic benefit of integrating PV and battery storage into Johannesburg’s commercial energy landscape.

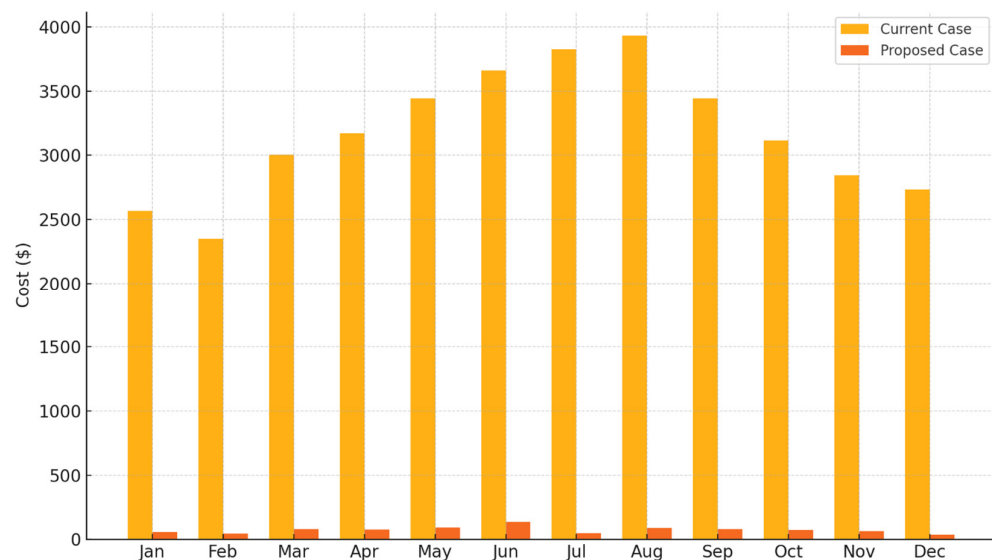


Figure 5. Monthly utility bill comparison between current and proposed systems.

Monthly utility costs in the proposed system are consistently suppressed due to the PV and storage capacity offsetting grid reliance. July and August mark the peak grid usage in the base case, which is nearly eliminated post-implementation. This facility uses 839 kWh/day and has a peak of 178 kW. In the proposed system, the following generation sources that serve the electrical load are illustrated in Figure 6.

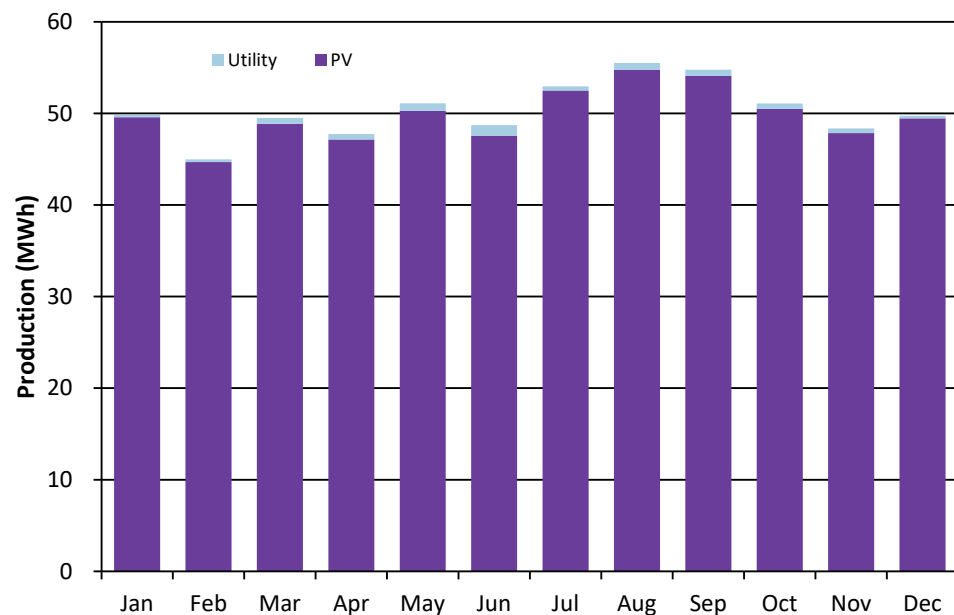


Figure 6. Monthly electric consumption.

3.2. Cash Flow

We evaluated the annual financial impact through a cash flow analysis of deploying the proposed hybrid system over a 25-year project lifespan. This includes capital and operational expenditures, tax benefits, and the net impact of utility expenses, adjusted for an 8.0% nominal discount rate, 2.0% inflation, and a real interest rate of 5.9%. The cash flow structure incorporates deductions such as Bonus Depreciation and Modified Accelerated Cost Recovery System (MACRS), alongside expenditures for battery maintenance, PV servicing, and residual utility costs under a simplified tariff. Notably, no Investment Tax Credits were assumed for the evaluation period. Operationally, the generic 1 kWh Li-Ion [ASM] system incurs a recurring cost of approximately \$270/year, while the flat-plate PV system requires about \$26/year in maintenance. Despite these, significant annual savings are realized through drastically reduced utility charges—approximately \$583/year—replacing the prior grid-heavy expenses.

The inclusion of MACRS in the first six years yields accelerated tax deductions, particularly prominent in years 1–3. These tax benefits contribute to a more favorable net present cost and improve internal rate of return for the project. The annual cash flow summary is tabulated in Table 4.

Table 4. Annual Cash flow summary.

Year	Bonus Depreciation	Li-Ion ASM (\$)	Flat-Plate PV (\$)	Investment Tax Credit	MACRS (\$)	Simple Tariff (\$)
1	\$0.00	(\$270.00)	(\$25.80)	\$0.00	\$749.31	(\$582.24)
2	\$0.00	(\$270.00)	(\$25.80)	\$0.00	\$1199.40	(\$582.24)
3	\$0.00	(\$270.00)	(\$25.80)	\$0.00	\$719.67	(\$582.24)
4	\$0.00	(\$270.00)	(\$25.80)	\$0.00	\$431.75	(\$582.24)
5	\$0.00	(\$270.00)	(\$25.80)	\$0.00	\$431.75	(\$582.24)
6	\$0.00	(\$270.00)	(\$25.80)	\$0.00	\$215.88	(\$582.24)
7–25	\$0.00	(\$270.00)	(\$25.80)	\$0.00	\$0.00	(\$582.24)

3.3. Performance Summary

Figure 7 illustrates distinct seasonal trends in system operation. Grid purchases peak in winter months (June–August) when solar irradiance is lowest, while battery SOC patterns show deeper discharge cycles during this period to maintain supply. In contrast,

during summer (December–February), excess solar generation keeps SOC consistently high, minimizing grid dependency. These trends highlight the system’s dynamic response to seasonal load–supply imbalances and reinforce its role in flattening grid import peaks.

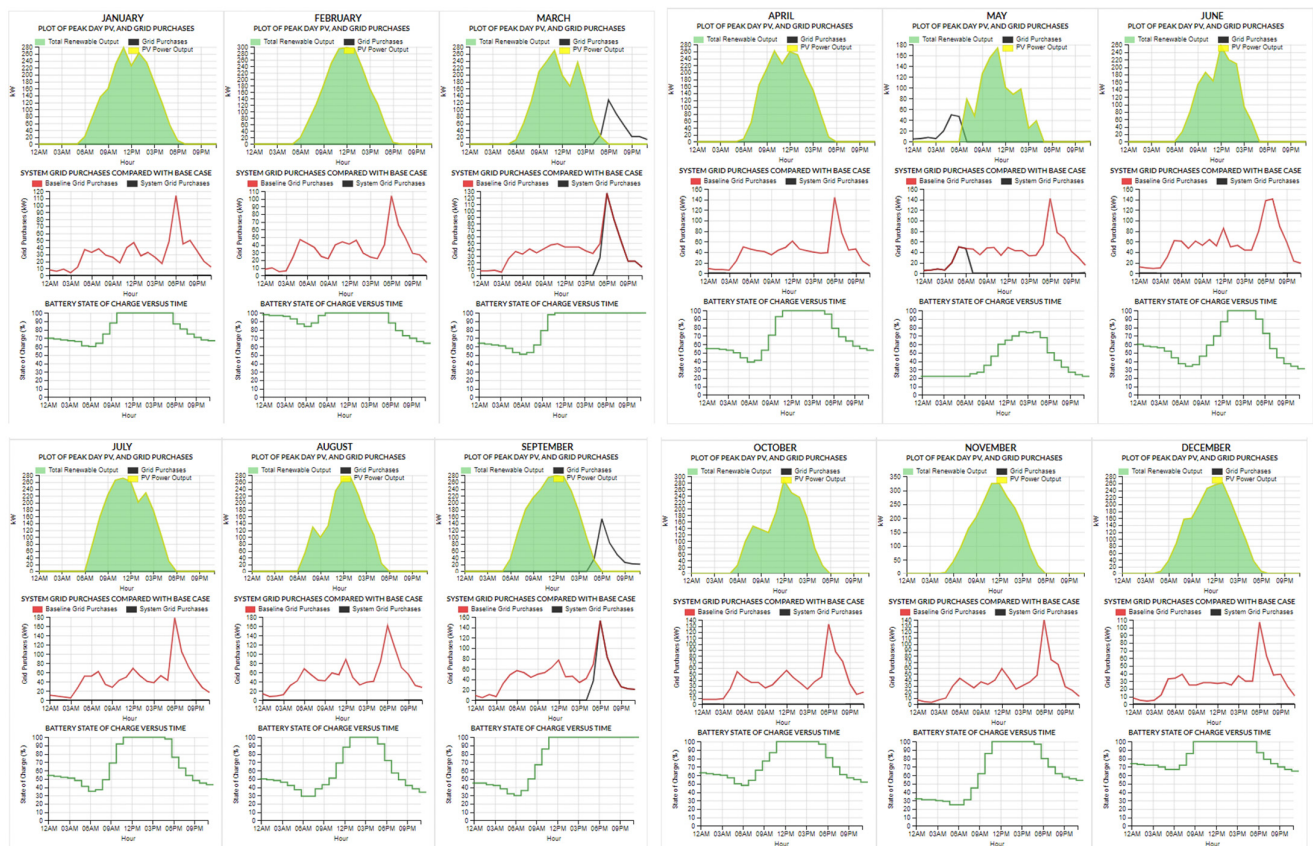


Figure 7. Monthly operational performance.

3.4. Environmental Impact—Carbon Emissions

Table 5 outlines the monthly carbon dioxide emissions for grid system, reflecting consistent reductions throughout the year. Emissions savings peak in January and December, with the system achieving a net annual reduction of 163 metric tons. Based on an average South African grid emission factor of 0.92 kg CO₂/kWh, this corresponds to an estimated 35–40% reduction in grid-related emissions for the modeled load. This percentage reflects a substantial decarbonization benefit given the country’s current coal-heavy energy mix. This performance highlights the system’s effectiveness in offsetting grid-based emissions through renewable generation and battery optimization.

Table 5. Monthly carbon dioxide emissions—grid system.

Month	Monthly CO ₂ Reduction (t)	Implied Energy Offset (MWh)
January	−17.0	−18.48
February	−15.0	−16.3
March	−14.0	−15.22
April	−12.0	−13.04
May	−12.0	−13.04
June	−9.2	−10.0
July	−11.0	−11.96
August	−13.0	−14.13

Table 5. Cont.

Month	Monthly CO ₂ Reduction (t)	Implied Energy Offset (MWh)
September	−15.0	−16.3
October	−14.0	−15.22
November	−15.0	−16.3
December	−16.0	−17.39
Annual Total	−163.2 t/yr	−177.39 MWh/yr

Column 3 presents the implied energy offsets, derived using South Africa’s grid emission factor to contextualize the reported CO₂ reductions. The summary of monthly CO₂ reductions indicates improved environmental performance with peak offsets in winter and consistent savings year-round.

3.5. Sensitivity Analysis—Impact of Solar Irradiance

Figure 8 illustrates the effect of varying global solar irradiance on the system’s NPC, LCOE, and annual operating cost. As solar radiation increases from 4.5 to 9.0 kWh/m²/day, all three metrics decline, indicating improved system efficiency and reduced reliance on grid imports. The sharp decrease in NPC between 4.5 and 7.0 kWh/m²/day reflects the system’s high sensitivity to lower irradiance levels, with cost-saving benefits tapering beyond 7.0 kWh/m²/day. These results confirm the economic competitiveness of PV deployment in high-irradiance urban environments like Johannesburg.

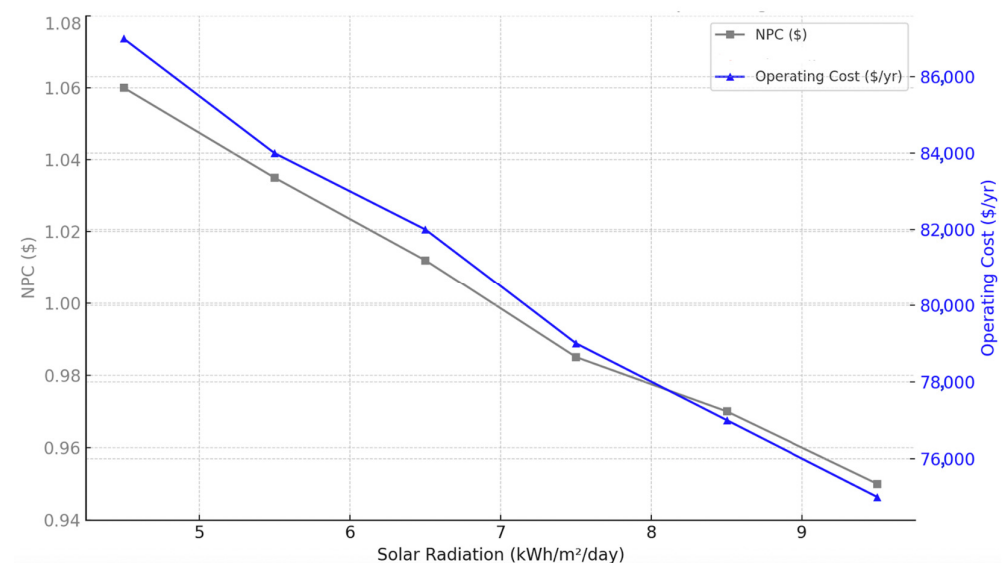


Figure 8. Sensitivity analysis—impact of solar irradiance.

These findings align with prior techno-economic assessments in high-solar regions such as Kaduna, Nigeria [25], and Riyadh, Saudi Arabia [24], where optimized PV/battery configurations yielded similar or slightly higher payback periods (1.1–1.5 years) and LCOE ranges between \$0.04–\$0.07/kWh. Comparing these to the current study’s LCOE of \$0.0053/kWh and payback period of less than two years, the system proposed here demonstrates superior performance—largely due to Johannesburg’s strong solar resource and lower capital cost assumptions. However, these results are based on modeled assumptions without real-world implementation and, thus, should be interpreted with an awareness of possible deviations due to equipment aging, tariff changes, and unmodeled grid constraints.

Table 6 summarizes the LCOE values from comparable studies in similar solar-rich regions, confirming the proposed system’s competitive economic performance.

Table 6. Comparison of LCOE with similar studies in high-solar regions.

Location	Climate Type	System Type	LCOE (\$/kWh)	Reference
Johannesburg, SA	Subtropical Highland	PV + Battery (Grid-tied)	0.0053	This Study
Kaduna, Nigeria	Tropical Savanna	PV + Battery (Grid-tied)	0.040–0.065	[25]
Riyadh, Saudi Arabia	Hot Desert (BWh)	PV + Battery	0.0637	[24]
Dhaka, Bangladesh	Humid Subtropical	PV + Battery	0.052	[12]
Giri Town, Nigeria	Tropical Monsoon	PV + Battery	0.038–0.060	[26]
Sri Lanka (Peri-Urban)	Tropical Rainforest	PV + Battery + Diesel	0.034	[27]

To assess the financial resilience of the proposed hybrid PV/BESS system under varying macroeconomic conditions, an additional sensitivity analysis was conducted on three critical economic parameters: discount rate, inflation rate, and battery replacement cost. These factors are particularly relevant in the South African context, where fiscal volatility, policy uncertainty, and inflationary pressures can influence long-term investment outcomes. As shown in Figure 9, all three variables exhibit a measurable impact on the system’s NPC. An increase in the real discount rate from 6% to 12% raises the NPC from approximately \$0.96 million to \$1.04 million, while inflation rate variations from 2% to 6% shift the NPC from \$0.98 million to \$1.05 million. The system is most sensitive to battery replacement cost escalations, with a linear increase in NPC from \$0.80 million to \$1.20 million as replacement costs rise by 50% increments. These results highlight the importance of robust financial planning and underscore the system’s vulnerability to capital cost fluctuations—particularly battery-related expenditures.

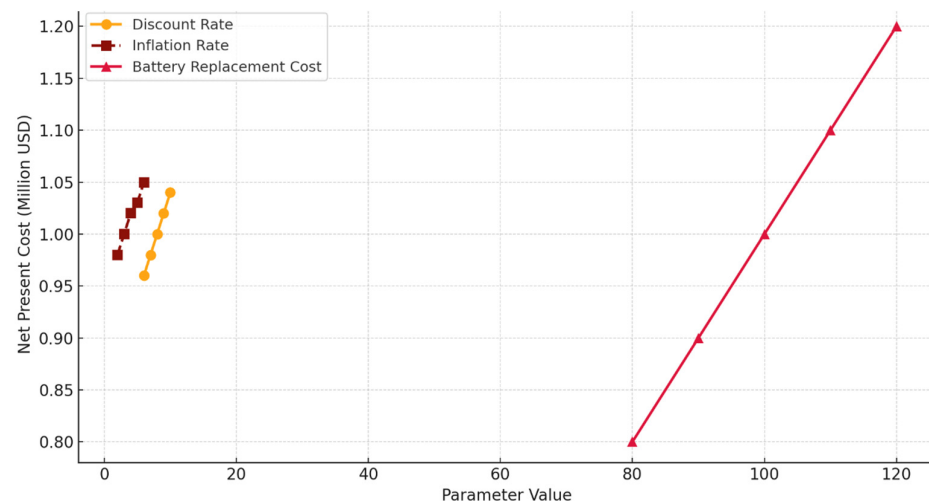


Figure 9. Sensitivity of NPC to economic parameters.

Battery degradation effects were implicitly captured by modeling a $\pm 20\%$ variation in replacement cost to reflect possible early replacements or reduced usable capacity due to cycling and aging. This sensitivity accounts for financial exposure under real-world performance conditions, aligning system economics with long-term operational risk.

While the proposed system performs well under typical operating conditions, real-world scenarios may introduce disruptions such as prolonged cloud cover, equipment downtime, or unanticipated demand spikes. Although not explicitly simulated, these atypical events could temporarily increase reliance on the grid or reduce storage efficiency.

Future work should include stochastic modeling or Monte Carlo simulations to assess system resilience under such non-ideal conditions.

4. Conclusions

This study demonstrates the strong techno-economic viability of deploying a grid-tied PV and lithium-ion battery storage system for a commercial facility operating under Johannesburg’s subtropical highland climate. Leveraging the city’s high solar irradiance and confronting the ongoing challenges of grid instability and tariff inflation, the proposed hybrid system achieves substantial electricity cost savings—reducing utility expenses by over 97% and delivering a simple payback period of less than one year.

Through detailed simulations in HOMER Grid and a rigorous sensitivity analysis, the system is shown to be both financially sound and operationally resilient, with an NPV of approximately \$449,491 over 25 years. Beyond cost savings, the hybrid solution contributes meaningfully to sustainability objectives, offsetting 163 metric tons of CO₂ annually, and enhancing energy autonomy in a context where power reliability remains a critical issue. This research confirms the potential of urban grid-tied PV/BESS configurations as a strategic solution for commercial energy consumers in South Africa. As utility-scale grid reforms continue and energy policy evolves, such decentralized systems provide an immediately deployable, scalable, and climate-aligned path forward for urban energy resilience and economic sustainability.

While the system demonstrates strong techno-economic performance in simulation, the results are based on modeled assumptions; future work should incorporate empirical validation through pilot projects and explore risks related to system degradation, cost shifts, and policy changes.

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Abbreviations

The following abbreviations are used in this manuscript:

Symbol/Abbreviation	Description	Unit
P _{PV}	Photovoltaic	—
BESS	Battery energy storage system	—
NPC	Net present cost	USD
NPV	Net present value	USD
LCOE/COE	(Levelized) Cost of Energy	USD/kWh
SOC	State of charge	%
HOMER	Hybrid optimization of multiple energy resources	—

ASM	Advanced storage module	—
Gt	Global solar radiation at time t	kW/m ²
APV	PV array area	m ²
η PV	PV conversion efficiency	%
E _{served}	Annual energy delivered to the load	kWh
YPV	PV capacity at Standard Test Conditions (STC)	kW
fPV	PV derating factor	%
GT	Global solar irradiance	kW/m ²
TC	PV cell temperature	°C
α P	Temperature coefficient of PV power	%/°C
E _{max}	Nominal battery energy capacity	kWh
P _{load}	Electrical load demand	kW
PPV	PV output power	kW
P _{batt}	Battery charge/discharge power	kW
P _{grid}	Grid import/export power	kW
η c, η d	Charge/discharge efficiency	%
Δt	Simulation time step	h
i	Real discount rate	%
n	Project lifetime	years
CRF	Capital recovery factor	—
TAC	Total annualized cost	USD/year
MACRS	Modified Accelerated Cost Recovery System	—
ITC	Investment Tax Credit	—
kWh	Kilowatt-hour (energy)	kWh
kW	Kilowatt (power)	kW
CO ₂	Carbon dioxide	metric tons or kg

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